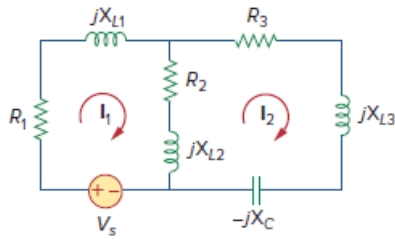


Problem

Using Fig. 10.77, design a problem to help other students better understand mesh analysis.

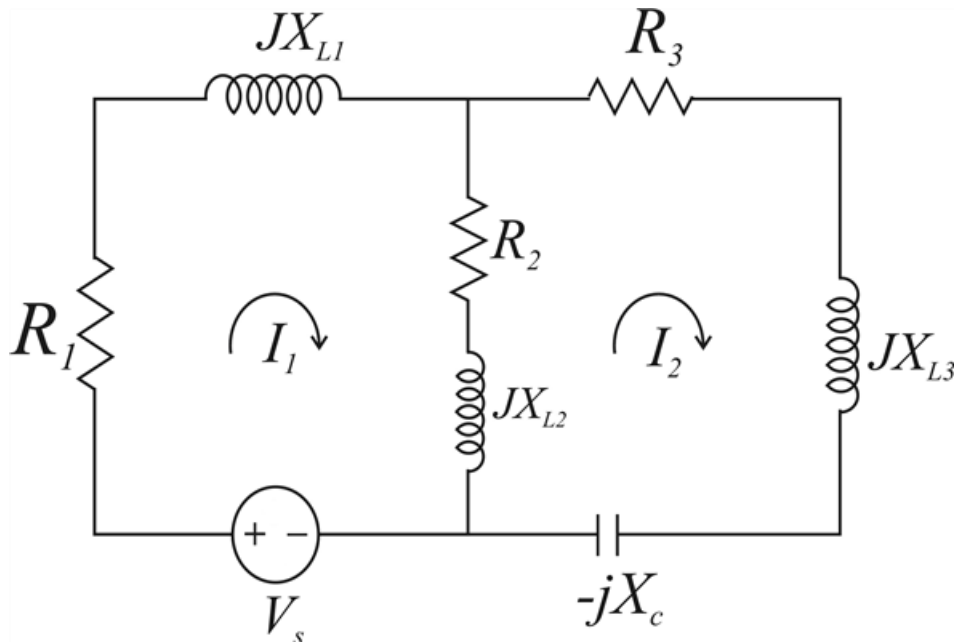
Figure 10.77:



Step-by-step solution

Step 1 of 3

Given circuit is,



Step 2 of 3

Applying KVL to mesh 1

$$-V_s + I_1 R_1 + jX_{L1} I_1 + R_2 (I_1 - I_2) + jX_{L2} (I_1 - I_2) = 0$$

$$(R_1 + R_2 + j(X_{L1} + X_{L2})) I_1 - (R_2 + jX_{L2}) I_2 = V_s$$

Applying KVL to mesh2,

$$jX_{L2} (I_2 - I_1) + R_2 (I_2 - I_1) + R_3 I_2 + jX_{L3} I_2 + (-jX_C) I_2 = 0$$

$$-(R_2 + jX_{L2}) I_1 + (R_2 + R_3 + j(X_{L2} + X_{L3} - X_C)) I_2 = 0 \rightarrow (2)$$

Equations (1) and (2) can be put in matrix form as,

$$\begin{bmatrix} R_1 + R_2 + j(X_{L1} + X_{L2}) & -(R_2 + jX_{L2}) \\ -(R_2 + jX_{L2}) & R_2 + R_3 + j(X_{L2} + X_{L3} - X_C) \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} V_s \\ 0 \end{bmatrix}$$

by using cramer rule,

$$\Delta = \begin{vmatrix} R_1 + R_2 + j(X_{L1} + X_{L2}) & -(R_2 + jX_{L2}) \\ -(R_2 + jX_{L2}) & R_2 + R_3 + j(X_{L2} + X_{L3} - X_C) \end{vmatrix}$$

$$\Delta = R_1 R_2 + R_1 R_3 + R_2 R_3 + jR_1(X_{L2} + X_{L3}) + jR_2(X_{L1} + X_{L3}) + jR_3(X_{L1} + X_{L2}) - jX_L(R_1 + R_2) - X_{L1}(X_{L2} + X_{L3}) + X_C(X_{L1} + X_{L2}) - (X_{L2}(X_{L3}))$$

$$\Delta_1 = \begin{vmatrix} V_s & (R_2 + X_{L2}) \\ 0 & R_2 + R_3 + j(X_{L2} + X_{L3} - X_C) \end{vmatrix}$$

$$\Delta_1 = V_s(R_2 + R_3 + j(X_{L2} + X_{L3} - X_C))$$

$$\Delta_2 = \begin{vmatrix} R_1 + R_2 + j(X_{L1} + X_{L2}) & V_s \\ -(R_2 + jX_{L2}) & 0 \end{vmatrix}$$

$$\Delta_2 = V_s(R_2 + jX_{L2})$$

Therefore,

$$I_1 = \frac{\Delta_1}{\Delta}$$

$$I_2 = \frac{\Delta_2}{\Delta}$$